Bicycles Relocation

Mathematical Model Implementation

RentalBike

1 Instructions

- It is strongly advised that you write down any additional hypotheses that you feel are necessary for a good understanding of your answers.
- You can answer this test in either English, Portuguese or Spanish.

2 Challenge: Bicycles Relocation

2.1 Introduction

RentalBike is a bike sharing company specialized in short-term bicycle rentals along various areas of a city, offering different categories of bicycles. Clients may rent bicycles for any duration and may deliver the rented bicycles in a place different from where the rental started. However, the delivery place must be in one of the areas currently operated by RentalBike.

Since clients are able to change the bicycles' original position, RentalBike must often redistribute the surplus of bicycles according to the expected demand of each area. The decision of how many bicycles will be redistributed from one area to another is based on expected profits for each possible volume of bicycles to be moved.

Therefore, before starting the relocation of bicycles, it is necessary to optimize the volume of bicycles to be moved from one area to the others. The following sections describe the problem to be solved. Read the input descriptions and assumptions before doing the task.

2.2 Input Description

Given as input:

- a surplus of bicycles of different categories in the source area;
- constants for the space occupied by one bicycle of each category in the truck used for relocating different categories of bicycles;
- and the expected profits for relocating bicycles from the source area to any other area.

Your challenge is to determine how many bicycles of each category should be moved to each area so that the sum of expected profit is maximized.

2.3 Assumptions

1. The expected profits are constants regarding each unit of bicycles that can be relocated to other areas. Therefore, it is a problem of deterministic optimization;

2. In this challenge, only one area is the source of bicycles. Therefore, there will be only relocations starting from this area.

The input is sourced as one workbook file named `BicyclesRelocationData.xlsx` with spreadsheets described below:

- A spreadsheet named "Categories" has each bicycle category as header, and respectively, two rows of data:

  1. the surplus of bicycles available in the source area for each category;
  2. and the constant for how much space each bicycle category occupies in the truck.

  As an example, if a bicycle category named "Child" has constant 1 and the category "Adult" has constant 1.5, then an "Adult" bicycle occupies 50% more space in the truck than a "Child" bicycle.

- Spreadsheets with names starting with "ExpectedProfitsArea" and ending with a number. These numbers indicate which destination area is the spreadsheet describing. Each of these spreadsheets has all bicycle categories as header and various rows of data. Each row presents how much profit it is expected to be earned by sending one additional bicycle to that destination area.

2.4 Characteristics of the Expected Profit

- The expected profits data is **not accumulated**. As an example, the expected profit of the second rental is only about the second rental. If you need the expected profits, for example, for renting 15 bicycles of a category in a specific area, it is needed to sum all the first 15 values regarding this bicycle category and area;

- Since the bike sharing demand varies according to different factors, the profit earned by the first rental of a bicycle may be greater or lower than the profit earned by the second rental of the same bicycle in the same region;

- The expected profit for relocating the n^{th} bicycle is **only earned if all earlier bicycles** of this same category and area are rented. In other words, the 30^{th} expected profit of a bicycle category and area is only earned if its earlier 29 bicycles were already relocated.

- The only property the expected profit values have is that they are **non-negative numbers**. Therefore, it is not possible to make other assumptions about the series of expected profits, such as linear or convex growth;

- The length of the expected profits data may be different than the surplus of bicycles of each category. In the case of not having expected profits data for all surplus bicycles, the profit for relocating more bicycles than the category surplus must be considered as **zero**.

3 Tasks

This challenge is divided into minor tasks, described as follows.

1. **Mathematical Formulation:** Write a mathematical formulation for solving this optimization problem, considering that the total space capacity of the truck is represented by a constant T. Define the constants and variables, and describe the constraints and objective function to maximize the sum of expected profits.

2. **Implementation and Solution:** Choose a programming language and a solver (such as CBC or GLPK) for optimizing the formulation that you propose at task 1. Considering the capacity of the truck as $T = 80$. Solve the problem and show your code, the optimal solution, and the optimal objective value.

4 Additional Requirements - AI Tool Usage

If you used any AI tools (ChatGPT, Copilot, etc.), document clearly:

- Which parts of the code were AI-assisted.
- How you validated the AI-generated code.
- Your reflection on the effectiveness of using AI for this task.

Include this information in a separate section of your submission.

5 Submission

Please submit a single compressed file (ZIP or TAR.GZ) containing:

- Source code directory with all implementation files.
- README file with setup and execution instructions.
- Results with the best found solution and its objective value.

Name the file: `bicycles_relocation_exact_<your_name>.zip`

6 Final Remarks

- There is no single "correct" answer — we value your reasoning process.
- Questions and requests for clarification can be sent by e-mail.
- The technical interview will focus on discussing your implementation and design choices.

Good luck!